

Mathematical Statistics

Lecture 7: Random Variables and Probability Distributions

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1 Introduction: What is a Random Variable?

The concept of a **random variable** is one of the most fundamental building blocks in probability and statistics. It allows us to connect the outcomes of a random experiment to a numerical value, which we can then analyze mathematically.

- **Analogy to Algebra:** In algebra, a variable like x is a placeholder for a number (e.g., in the equation $x + 5 = 8$, x takes the value 3).
- **The "Random" Element:** A random variable works similarly, but it represents the numerical outcome of a random process. Before the process happens, we don't know the exact value of the variable, but we can determine the *set of possible values* it can take and the *probability* of it taking each of those values.

💡 Theoretical Foundations

A **random variable** is a variable whose value is a numerical outcome of a random phenomenon. It is a function that maps each outcome in a sample space (Ω) to a real number. We typically denote random variables with capital letters (e.g., X, Y, Z).

2 Types of Random Variables

Random variables can be broadly classified into two main types based on the values they can assume.

2.1 Discrete Random Variables

A random variable is **discrete** if it can take on a finite or countably infinite number of distinct values. You can often think of these as values you can "count."

Examples:

- The number of heads obtained in 3 coin flips (can be 0, 1, 2, or 3).
- The result of a single roll of a six-sided die (can be 1, 2, 3, 4, 5, or 6).
- The number of defective items in a sample of 20 products.
- The number of emails you receive in an hour.

2.2 Continuous Random Variables

A random variable is **continuous** if it can take on any value within a given range or interval. These are values you typically "measure."

Examples:

- The height of a person (can be 1.75m, 1.751m, 1.7511m...).
- The temperature of a room.
- The time it takes to complete a task.
- The exact weight of an orange.

3 Probability Distributions for Discrete Random Variables

Once we have a random variable, our next step is to describe how likely each of its possible values is. This is done using a **probability distribution**. For discrete variables, this is called a **Probability Mass Function (PMF)**.

3.1 The Probability Mass Function (PMF)

The PMF, denoted as $P(X = x)$, gives the probability that the random variable X takes on a specific value x .

3.1.1 Key Properties of a PMF

1. **Non-Negativity:** The probability for any value x must be non-negative.

$$P(X = x) \geq 0 \quad \text{for all possible values of } x$$

2. **Sum to One:** The sum of the probabilities over all possible values of x must equal 1.

$$\sum_{\text{all } x} P(X = x) = 1$$

3.2 Core Example: Flipping a Coin 4 Times

Let's build a PMF from a classic experiment.

- **Experiment:** Flip a fair coin 4 times.
- **Sample Space (Ω):** The set of all possible outcomes. There are $2^4 = 16$ outcomes.

$$\Omega = \{HHHH, HHHT, \dots, TTTT\}$$

- **Random Variable (X):** Let X be the **number of heads** that occur.
- **Possible Values for X:** X can take values from the set $\{0, 1, 2, 3, 4\}$.

3.2.1 Step-by-Step Calculation of the PMF

Using combinations ($\binom{n}{k}$), we find the number of ways to get k heads in $n = 4$ flips.

- $P(X = 0)$ (0 heads, e.g., TTTT): $\frac{\binom{4}{0}}{16} = \frac{1}{16}$
- $P(X = 1)$ (1 head, e.g., HTTT): $\frac{\binom{4}{1}}{16} = \frac{4}{16}$
- $P(X = 2)$ (2 heads, e.g., HHTT): $\frac{\binom{4}{2}}{16} = \frac{6}{16}$
- $P(X = 3)$ (3 heads, e.g., HHHT): $\frac{\binom{4}{3}}{16} = \frac{4}{16}$
- $P(X = 4)$ (4 heads, e.g., HHHH): $\frac{\binom{4}{4}}{16} = \frac{1}{16}$

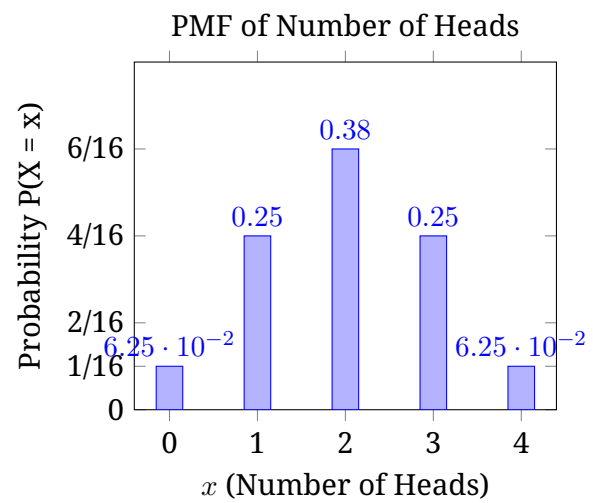
This particular distribution is a **Binomial Distribution**.

3.2.2 Tabular and Visual Representation of the PMF

PMF for Number of Heads in 4 Coin

	x (Heads)	P(X = x)
Flips	0	1/16
	1	4/16
	2	6/16
	3	4/16
	4	1/16

Check: $\frac{1}{16} + \frac{4}{16} + \frac{6}{16} + \frac{4}{16} + \frac{1}{16} = \frac{16}{16} = 1$.
The property holds.



4 Worked Examples: Solving Problems with PMFs

4.1 Example 1: Finding an Unknown Constant 'k'

A random variable X has the PMF: $P(X = x)$ is 0 for $x = 0$; k for $x = 1$; $2k$ for $x = 2$; $2k$ for $x = 3$; $3k$ for $x = 4$; k^2 for $x = 5$; $2k^2$ for $x = 6$; and $7k^2 + k$ for $x = 7$. Find k .

Solution: We use the property that $\sum P(X = x) = 1$.

$$0 + k + 2k + 2k + 3k + k^2 + 2k^2 + (7k^2 + k) = 1$$

Combine like terms:

$$(k^2 + 2k^2 + 7k^2) + (k + 2k + 2k + 3k + k) = 1$$

$$10k^2 + 9k = 1 \implies 10k^2 + 9k - 1 = 0$$

This is a quadratic equation. We solve by factoring:

$$10k^2 + 10k - k - 1 = 0 \implies 10k(k + 1) - 1(k + 1) = 0 \implies (10k - 1)(k + 1) = 0$$

This gives two possible solutions: $k = 1/10$ or $k = -1$. Since probability cannot be negative ($P(X = 1) = k \geq 0$), we discard $k = -1$.

Therefore, $k = 1/10$ or 0.1 .

4.2 Example 2: Probability with Combinations (The Bad Oranges)

A box contains 16 good and 4 bad oranges. Two oranges are drawn at random. Let X be the number of bad oranges drawn. Find the PMF of X .

Solution:

- Total oranges = $16 + 4 = 20$. Total combinations of drawing 2: $\binom{20}{2} = \frac{20 \times 19}{2} = 190$.
- X can take values $\{0, 1, 2\}$.
- $P(X = 0)$ (0 bad, 2 good): $P(X = 0) = \frac{\binom{16}{2} \times \binom{4}{0}}{\binom{20}{2}} = \frac{120 \times 1}{190} = \frac{120}{190} = \frac{12}{19}$.
- $P(X = 1)$ (1 bad, 1 good): $P(X = 1) = \frac{\binom{16}{1} \times \binom{4}{1}}{\binom{20}{2}} = \frac{16 \times 4}{190} = \frac{64}{190} = \frac{32}{95}$.
- $P(X = 2)$ (2 bad, 0 good): $P(X = 2) = \frac{\binom{16}{0} \times \binom{4}{2}}{\binom{20}{2}} = \frac{1 \times 6}{190} = \frac{6}{190} = \frac{3}{95}$.

PMF for Number of Bad Oranges Drawn	x (Bad Oranges)	0	1	2
	P(X = x)	12/19	32/95	3/95

5 The Cumulative Distribution Function (CDF)

The PMF gives the probability of *exactly* one value. Sometimes, we want to know the probability of a random variable being *less than or equal* to a certain value. This is given by the **Cumulative Distribution Function (CDF)**, denoted $F(x)$.

Theoretical Foundations

Definition: The CDF is defined as $F(x) = P(X \leq x)$. It is the "running total" of the probabilities from the PMF.

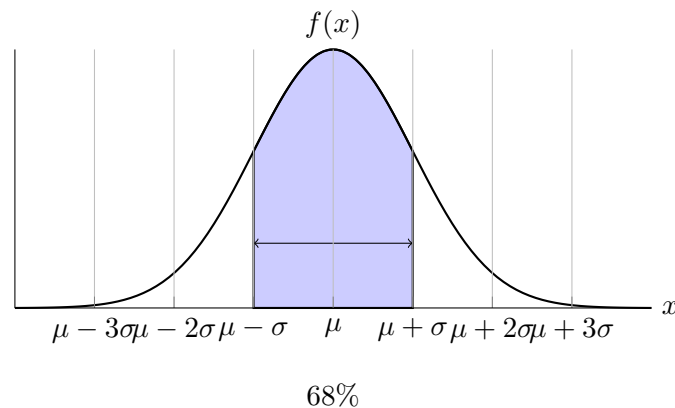
For the 4 coin flips example:

- $F(0) = P(X \leq 0) = P(0) = \mathbf{1/16}$
- $F(1) = P(X \leq 1) = P(0) + P(1) = 1/16 + 4/16 = \mathbf{5/16}$
- $F(2) = P(X \leq 2) = F(1) + P(2) = 5/16 + 6/16 = \mathbf{11/16}$
- $F(3) = P(X \leq 3) = F(2) + P(3) = 11/16 + 4/16 = \mathbf{15/16}$
- $F(4) = P(X \leq 4) = F(3) + P(4) = 15/16 + 1/16 = \mathbf{1}$

6 Probability Distributions for Continuous Variables

For continuous variables, the probability of X being *exactly* a specific value is zero. Instead, we use a **Probability Density Function (PDF)**, $f(x)$, where probability is the **area under the curve** between two points.

A classic example is the **Normal (Gaussian) Distribution**, often called the "bell curve."



- The total area under the PDF curve must equal 1.
- We can't ask "What is $P(\text{Height} = 6 \text{ feet})$ ". This is 0.
- Instead, we ask "What is $P(5.5 \leq \text{Height} \leq 6)$ ". This corresponds to the area under the curve between 5.5 and 6, calculated with an integral:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Why This Matters

1. **Concise Summary:** Distributions provide a simple way to summarize probabilities of complex experiments. Describing 100 coin flips (2^{100} outcomes) is impossible by hand, but the number of heads is easily described by a **Binomial Distribution**.
2. **Mathematical Models:** They provide functions (PMFs and PDFs) that act as models for real-world random processes. We don't have to count outcomes if we have a function.
3. **Foundation for Statistics:** They allow us to perform **hypothesis testing** (e.g., "Is this coin fair?") and **parameter estimation** (e.g., "What is the average height based on my sample data?").
4. **Enabling Powerful Computations:** We can use summation (discrete) and calculus (continuous) to calculate probabilities over ranges, find expected values (averages), and measure variance (spread).
5. **Predicting and Propagating Uncertainty:** We can model uncertainty (e.g., in a measurement) as a random variable and then see how that uncertainty propagates through a system or calculation. This is crucial in engineering, physics, and finance.

Theoretical Foundations

- A **random variable (X)** assigns a numerical value to each outcome of a random experiment.
- **Discrete** variables are countable (number of heads), while **continuous** variables are measurable (height).
- The **Probability Mass Function (PMF)** gives the probability $P(X = x)$ for each value of a discrete random variable. All probabilities must sum to 1.
- The **Probability Density Function (PDF)** describes the probability distribution for a continuous random variable, where probability is the area under the curve.
- The **Cumulative Distribution Function (CDF)** gives the cumulative probability $P(X \leq x)$.
- These tools are essential for modeling randomness and performing statistical analysis.

7 Practice Problems

1. A six-sided die is rolled twice. Let the random variable Y be the sum of the two rolls.
 - (a) What are the possible values for Y ?
 - (b) Find the PMF of Y . (Hint: Create a 6x6 grid of outcomes).
 - (c) What is $P(Y > 9)$?

2. The number of customer complaints received by a company per day is a random variable X with the following PMF:

$$P(X = x) = k \cdot (x + 1) \quad \text{for } x = 0, 1, 2, 3.$$

- (a) Find the value of the constant k .
 - (b) What is the probability that the company receives at most 2 complaints in a day, i.e., $P(X \leq 2)$?
3. Classify the following random variables as **discrete** or **continuous**:
- (a) The number of cars that pass through an intersection in an hour.
 - (b) The exact time it takes to run a marathon.
 - (c) The number of questions a student answers correctly on a 20-question test.
 - (d) The volume of water in a 1-liter bottle.